

Goertzel 4CP v23.5: Residual Boundary Peeling after the IDBRAD Obstruction

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May 8, 2026

Abstract

This memo records a standalone v23.5 repair attempt for the Goertzel four-color route. The exact v23 Step 2 purification statement remains valid in Lean: single-component purification holds, and summing those local identities recovers the boundary-only generator used downstream. What fails is the present ambient interior-dual boundary-root adjacent-distance peeling package: on honest closed-walk geometry with genuine interior support, Lean proves that the current IDBRAD surface cannot exist. The proposed repair is therefore not to weaken or patch away that obstruction, but to move the induction to a residual/current-boundary peeling lane whose certificate is tied to the live boundary created by previous cancellations. This document is a route specification and proof attempt, not a proof of the Four Color Theorem. Lean now also proves that the current residual/current-boundary positive wrapper is equivalent, on a fixed embedding, to the existing collar-layer and height-ordered positive surfaces. So the present v23.5 work supplies a conservative reformulation and sharper obstruction tests, not a validated stronger same-embedding workaround.

1 Proved Lean facts

The current Lean status is already sharp enough to separate what survives from what is blocked. The main proved facts needed for a v23.5 memo are:

- `v23_single_component_purification` proves the exact v23 Step 2 local identity $X_f^{ab}(C) \oplus X_f^{ab}(C^D) = c \cdot 1_{D \cap \partial f}$ for one boundary-hitting (a, b) -Kempe component D .
- `sum_v23_single_component_purifications_eq_boundaryOnlyGenerator` shows that summing the exact Step 2 purifications over the boundary-hitting components recovers the boundary-only generator already used later in the algebraic route.
- `PlaneEmbeddingWithBoundary.FaceBoundaryClosedWalk.three_le_faceBoundary_card` shows that every honest nonempty simple facial closed walk has at least three boundary edges.
- `InteriorDualBoundaryRootAdjDistancePeelData.false_of_mem_interiorEdgeSupport_of_faceBoundaryClosedWalkGeometry` shows that any ambient IDBRAD package covering an interior edge would force an impossible terminal peeled face.
- `not_nonempty_interiorDualBoundaryRootAdjDistancePeelData_of_interiorEdgeSupport_nonempty_of_faceBoundaryClosedWalkGeometry` therefore rules out any nonempty ambient IDBRAD package on honest closed-walk geometry with genuine interior support.

- `nonempty_planarBoundaryResidualBoundaryLayerFacePeelWitnessData_`
`iff_nonempty_planarBoundaryCollarLayerFacePeelWitnessData,`
`theorem49ResidualBoundaryPositiveProjectedGeometryOn_iff_`
`collarLayerPositiveProjectedGeometryOn,` and `theorem49ResidualBoundaryPositiveProjectedGeomet`
`iff_heightOrderedPositiveProjectedGeometryOn` show that the current residual/current-
boundary positive wrapper has exactly the same fixed-embedding strength as the pre-existing
collar-layer and height-ordered witness surfaces.
- `nonempty_sharedInteriorPair_v23ResidualBoundaryInitialState_`
`sipFace0Boundary` builds the exact v23 residual initial state on a genu-
ine face boundary of the shared-interior-pair honest source benchmark, while
`sharedInteriorPair_closedWalkSource_tait_hasUnblockedInteriorEndpoint_`
and `v23ResidualBoundaryInitialState_without_residualBoundaryGeometry`
and `not_forall_residualBoundaryPositiveProjectedGeometryOn_of_`
`closedWalkAnnulusBoundarySource_and_v23ResidualBoundaryInitialState_and_`
`taiteEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` show
that even this exact Step 2 seed, together with honest source data, a Tait coloring, and
the purified endpoint witness, still does not force any residual/current-boundary positive
geometry on the same embedding.

The first two theorems live in `Mettapedia/GraphTheory/FourColor/GoertzelLemma44.lean`; the closed-walk cardinality lower bound lives in `Mettapedia/GraphTheory/WalkPlanarEmbeddingWithBoundary.lean`; and the IDBRAD obstruction theorems live in `Mettapedia/GraphTheory/FourColor/Theorem49PlanarBoundaryPeeling.lean`; the residual-equivalence theorems live in `Mettapedia/GraphTheory/FourColor/Theorem49ResidualBoundaryPeeling.lean`.

This fixes the status line exactly. Step 2 purification is not the live bug. The live bug is the ambient terminal-peeling interface currently encoded by IDBRAD. Any v23.5 route that tries to smuggle that same interface back in under weaker rhetoric should be rejected immediately by the existing regression theorems.

2 The v23.5 proof attempt

The strategic shift is simple to state and difficult to prove: stop asking the original ambient annulus boundary to own the whole induction. The present obstruction says that if every terminal peeled remainder must already sit on that ambient boundary, then genuine interior support forces an impossible one-edge face. That is valuable information. It tells us the induction parameter is wrong.

The v23.5 proposal is to let the algebra follow the *live* boundary of the surviving residual equation rather than the fixed ambient boundary of the full embedding. After each certified cancellation, one forms a new residual object $\mathcal{R}_k$ together with its current boundary B_k . The next peeling move is certified only relative to B_k , not relative to the initial annulus boundary chosen before any cancellation happened.

In route form, the intended proof attempt is:

1. Start from the honest closed-walk source and the exact v23 Step 2 purification identity already proved in Lean.

2. Package the surviving algebra as a residual equation object $\mathcal{R}_0$ with a current boundary B_0 and explicit invariants saying which terms are still live.
3. Prove that any nonzero residual $\mathcal{R}_k$ exposes a current-boundary certificate: some face, component, or boundary-supported algebraic event that can be canceled without leaving the residual framework.
4. Produce the next residual $\mathcal{R}_{k+1}$, with a strictly smaller measure, and repeat.
5. Reach contradiction only by exhausting the residual, not by demanding in advance that the whole ambient embedding already carries an IDBRAD-style terminal ownership package.

The creative claim of v23.5 is therefore not that the present IDBRAD obstruction is illusory. It is that the obstruction identifies the wrong globalization step. The proof route should chase the moving frontier created by surviving source terms, not the frozen frontier of the original ambient annulus. If that stronger live-boundary certificate exists, then the exact Step 2 purification theorem becomes the correct algebraic entry point instead of a local identity waiting for an impossible ambient peel theorem.

The central missing theorem is now clear: every nonzero residual satisfying the live-boundary invariants must admit a boundary-aligned algebraic cancellation certificate on the current boundary. Without that theorem, residual current-boundary peeling is only a plausible narrative. With it, one has a credible replacement for the blocked ambient lane.

The new shared-interior-pair calibration makes this sharper. The issue is not that Lean lacks the exact v23 Step 2 seed object: that seed already exists on a real source face boundary. What is missing is the genuinely geometric bridge from that algebraic seed to any same-embedding residual/current-boundary witness surface.

Lean now shows this already at the actual boundary-order source shell, not only after lowering to an honest closed-walk package. On the shared-interior-pair benchmark, explicit annulus boundary reachability, successor-cycle selected boundary arcs, the concrete Tait coloring, ‘HasUnblockedInteriorEndpoint’, and the exact v23 residual seed still do not force residual positive geometry. So the live missing theorem is genuinely route-facing source-to-residual geometry.

Lean now also shows that this exact v23 seed does not recover any of the older same-embedding geometry ladders either. On the same shared-interior-pair benchmark, the exact seed plus honest source data, the concrete Tait coloring, and ‘HasUnblockedInteriorEndpoint’ still do not force residual witness data, residual selector data, height-ordered witness data, collar-layer witness data, the attached boundary-rooted certificate, well-founded witness data, or weak annulus-collar geometry. The same failed-universal statement already holds on the actual successor-cycle boundary-order shell. So the remaining v23.5 burden is not a repackaging back into one of the existing geometry interfaces; it is a genuinely new source-to-residual bridge.

The current ambient IDBRAD obstruction should remain in place as a regression guard. A future proof that accidentally reconstructs ambient IDBRAD on honest closed-walk geometry has not repaired the route; it has walked back into a formally refuted interface.

3 Formal obligations

To turn the v23.5 story into Lean work, the roadmap is:

1. **Residual equation object.** Define a structure carrying the live generator, current support, current boundary, and the invariants preserved by one cancellation step.

2. **Live-boundary terminal theorem.** Prove that a nonzero residual cannot become terminal without exposing a boundary event on the current boundary. This replaces the ambient terminal-face claim used by IDBRAD.
3. **Algebraic cancellation certificate.** This is the central new obligation: from a live residual, extract a certified boundary-supported cancellation move that preserves honesty of the residual package.
4. **Residual annihilator theorem.** Show that repeated certified cancellations strictly decrease a well-founded measure and force the residual to vanish.
5. **Source construction theorem.** Build the initial residual object from the honest closed-walk source, the exact Step 2 purification theorem, and the boundary-only generator already present downstream.
6. **Regression tests.** Preserve the current ambient IDBRAD obstruction, and add negative tests showing that the new lane never silently reintroduces boundary-free ambient peeling, source-free witness placement, or a fake terminal singleton-face endpoint.

In particular, the algebraic cancellation certificate is the theorem that makes or breaks the entire v23.5 proposal. The rest of the roadmap organizes around it.

4 Status

No proof of the Four Color Theorem is claimed here. What is presently supported is narrower and firmer: the exact v23 Step 2 purification statement is proved, and the current ambient IDBRAD route is blocked by proved obstruction theorems. The residual/current-boundary lane now has a formal wrapper layer, but Lean also proves that the present positive wrapper is conservative: it is equivalent to the existing collar-layer and height-ordered packages on a fixed embedding. So v23.5 is still a proposed repair specification, not a completed derivation or a validated stronger same-embedding construction theorem.

Accordingly, this memo should be read as a repaired route document. It states what survives, what fails, and what new Lean obligations would have to be discharged before any proof claim could responsibly be made.